BENJAMIN FROHMAN

Lewisville / Frisco, Texas · Remote-ready · Available for research staff, advisor, and project-lead roles

Research systems design · Latent geometry & hierarchical representations · Multi-model coordination · Formal methods

SUMMARY

Independent researcher who designs complete, engineerable systems for frontier model training and coordination. Primary focus is the geometry of latent spaces, hierarchical (second-order) representations, and the infrastructure required for groups of models to share structured novelty signals under tight compute budgets. Delivers full technical proposals—schemas, phased roadmaps, risk analysis, and acceptance criteria—that research and infrastructure teams can act on directly. Combines geometric analysis, formal rigor, and systems thinking to produce work that is both conceptually deep and operationally usable. Accountable for clarity, constraints, and downstream executability.

AREAS OF FOCUS

Latent-space geometry and hierarchical representations · Multi-model coordination and novelty-signal exchange · Second-order models over latent statistics · Ontology and geometric classification of representation regions · Persistent homology and topological descriptors for high-dimensional data · Research infrastructure design (extraction, buses, feedback controllers, registries) · Pre-training and post-training systems concepts · Evaluation and synthetic-data prioritization · Formal methods and Lean 4 · End-to-end ownership of research agendas

SELECTED RESEARCH SYSTEMS & TECHNICAL WORK

Hierarchical Latent Exchange & Coordination Fabric (HLEX)

Independent systems design and technical report · 2026 · Draft technical proposal (v1.1)

- Designed a five-layer coordination fabric that allows a closed fleet of models under unified control to exchange rare-token novelty markers and compact embedding sketches, with concurrent co-occurrence treated as a first-class training and routing signal.
- Specified Marker Extraction Layer (with <1 % training overhead target), Shared Latent Signal Bus, Second-Order Model Service operating purely on latent-derived statistics, Feedback & Routing Controller, and Ontology & Geometry Registry.
- Produced complete Phase 0–3 roadmap with detailed workstreams, acceptance criteria, risk register, success metrics, and an appendix on persistent homology as a geometric classification tool for latent regions.
- Framed the design for direct use by research and infrastructure teams at frontier labs; document intended as both technical contribution and evidence of systems-level research leadership.

Geometric & Hierarchical Frameworks for High-Dimensional Structure

Independent research program · Ongoing · Multiple internal manuscripts and formalization efforts

- Developed hierarchical and dual geometric constructions intended to reorganize the analysis of complex high-dimensional dynamical systems into more tractable objects—work that directly informs the second-order and Registry components of HLEX.
- Advanced Lean 4 formalization pipelines and rigorous manuscript assembly processes, emphasizing machine-checkable claims, non-circular reasoning, and first-principles derivation standards.
- Applied topological and multi-scale geometric descriptors to characterize organization that remains invisible to purely local analytic methods; these descriptors appear as candidate geometric classifications inside the HLEX Registry.

Research Process & Tooling Systems

Independent development · 2025–2026

- Designed and iterated multi-stage research workflows for large technical manuscripts, formal verification, and systems proposals, including versioned archives, structured review cycles, and high-signal documentation standards.
- Built and refined personal research infrastructure supporting long-horizon technical projects under real resource and time constraints, with emphasis on reproducibility, auditability, and clear hand-off to other contributors.

PROFESSIONAL EXPERIENCE & OWNERSHIP

Independent Research Lead

Self-directed · 2024 – Present

- Own the full cycle of research systems work: problem selection, geometric method development, architecture, formal documentation, risk analysis, and production of stakeholder-ready technical proposals.
- Deliver complete, reviewable artifacts (schemas, roadmaps, acceptance criteria, appendices) that downstream engineering and research teams can execute without requiring the original author to remain the bottleneck.
- Maintain high accountability for both conceptual depth and operational constraints (compute overhead, multi-tenancy, phased adoption, security posture).

Progressive Insurance

Technical & operational roles · Ongoing

- Sustained professional delivery and accountability while simultaneously advancing independent research programs and degree completion focused on mathematics and geometric analysis.

EDUCATION & DEVELOPMENT

University of Texas at Dallas — Undergraduate studies in progress, mathematics focus, driven by long-term commitment to geometric analysis, rigorous foundations, and teaching. Extensive self-directed work in geometric analysis, topological data analysis, formal verification (Lean 4), large-model training systems concepts, and research infrastructure design. Continuous production of technical writing and design documents intended for research leadership review.

Full technical proposals, including HLEX Systems Design Proposal v1.1, available on request. Open to remote-friendly research staff, research advisor, and project-lead engagements that value both geometric depth and executable systems design.